An educational support tool based on robotic assistants, mobile apps, and expert systems for children with Down syndrome

V. Robles-Bykbaev, E. Andrade-Prieto, P. Solórzano-Guerrero, Y. Robles-Bykbaev and F. Pesántez-Avilés
GI-IATa, Cátedra UNESCO Tecnologías de Apoyo para la Inclusión Educativa
Calle Vieja 12-30 y Elia Liut, Cuenca, Ecuador
Email: vrobles@ups.edu.ec, eandradep@est.ups.edu.ec, pamesolorzanog@gmail.com, {zrobles, fpesantez}@ups.edu.ec

A. Parra-Astudillo
Inclutech-EC Inclusión y Tecnología S.A.
Calle Vega Muñoz y General Torres, Cuenca, Ecuador
Email: aparra@inclutechec.com

***Abstract*—The World Health Organization (WHO) claims that each year approximately 303,000 newborns die due congenital anomalies. These such of diseases can produce long-term disabilities, and the three most critical congenital anomalies are heart defects, neural tube defects, and Down syndrome (DS). Ecuador is a country with a high prevalence of DS, since 1 out of every 700 children born with this disease. On the other hand, worldwide, the number of children born with DS is 1 per 1,000. For these reasons, in this paper, we present a technological tool aimed at supporting the school curriculum for children from 6 to 12 years with DS. This tool relies on a low-cost robotic assistant, a mobile app, and a module to suggest educational activities for the children. With the aim of determining the children's response to the robot as well as to the mobile application, we have conducted a first pilot experiment with 21 children (without disabilities). Each of the participating children interacted with the system, and after this activity, they responded a survey. The initial results have shown that children are very interested and motivated to perform educational activities with the robot.**

***Keywords*—Down syndrome; Robotic assistant; Rule-based expert system; Mobile apps.**

I. Introduction

The Down Syndrome is produced by extra genetic material in chromosome 21. According to the WHO is a type of mental retardation in which the infants present the following physical characteristics: decreased muscle tone, a flat face, eyes slanting up, irregular shaped ears, ability to extend joints beyond the usual, etc. [1].

In Ecuador, the Solidary Mission "Manuela Espejo" of the Vice Presidency carried out a biopsychosocial-clinical-genetic study of people with disabilities. This study took more than a year in which a multidisciplinary team of Ecuadorian and international experts has visited the 24 provinces that constitute Ecuador. The results of this study have shown that 8,450 persons present disabilities related to chromosomal disorders. From this group, 7,557 persons have Down Syndrome distributed as follows: 48.24% are women whereas 51.76% are men [2], [3]. It is essential mentioning that severity of Intellectual Disability (ID) among individuals with DS falls on a spectrum that ranges from profound to borderline intellectual functioning (Intelligence Quotient [IQ] = 3070; mean IQ = 50) [4].

978-1-5386-5491-0/18/$31.00 ©2018 IEEE

Several studies have shown the benefits of including robotic assistants and technological support tools in education and care of children with different kinds of disabilities (intellectual, physical, visual, auditory, etc.). These tools can contribute towards eliciting motivation, concentration, taking initiative and improving attention span of the children when are appropriately integrated with standard rehabilitation protocols [5]–[7].

In the light of above, in this paper we present an educational support tool for children with DS. Our tool is aimed on supporting the school curriculum through interactive activities carried out with the support of a mobile application and a robotic assistant. In the same way, in our proposal we present the first stage of an expert system that relies on rules and has the objective of providing support to therapists/teachers for designing therapeutic and educational intervention plans. Our support tool covers the following educational areas:

- *Speech-Language Therapy (SLT):* children with DS have anatomical and physiological differences in the mouth and throat areas that make it more difficult for them to make precise movements [8]. For these reasons, our tool includes several activities to provide support in SLT.
- *Learning disabilities:* children with DS present difficulties in learning numbers compared to reading. In this area our system tries to provide support with the learning of numbers, daily activities (feeding, sleeping, etc.), spatial notions, etc.
- *Attention and concentration:* these children present attention deficit which can trigger inferior discrimination abilities. Therefore, the tool includes several mini-serious games to improve the attention span.
- *Body concepts and awareness:* in this area, the tool

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:52:37 UTC from IEEE Xplore. Restrictions apply.

provides activities to develop fine and gross motor skills as well as the internal understanding of where the body is in space.

II. Related work

In this section we will present some relevant proposals aimed at providing educational support for children/youth with DS and related disorders, such as mild ID or moderate ID. This brief overview will be focused on two kinds of educational support tools: mobile apps, and robotic assistants.

A. Mobile apps in education

In the area of mobile apps, the contributions are focused on supporting the development of different kinds of skills in persons with DS. Durango et al. conducted an experiment where they used mobile devices and serious games to develop the following skills in children with ages from 3 to 6 years: improve the care of children, working on the idea of similarities, improving visual memory, and improving/training vocabulary. The experiment was carried out with eight children with different kinds of difficulties or disabilities such as psychomotor retardation, DS, specific language impairment, premature birth, attention deficit disorder, and developmental delay. The authors report that 25% of children did not complete the activity when not using a digital game. In contrast, 100% of children were able to complete the activity using a digital game [9]. In a similar research line, Reis et al. present a case study of the use of mobile devices to support daily routines and promote digital literacy of a DS teen. The authors validated a conceptual model and of a prototype based on the instantiation of daily routine moments (pre-event, event and post-event) using the Unified Modeling Language (UML). To this aim, in this research was proposed a scenario organized in the three routine moments. For each moment the authors defined the following elements: user, requirements, and apps [10].

Similarly, Brooks presented a study where he examined the effects of self-delivered video prompts via an iPod Touch on teaching six adolescents with moderate to severe intellectual and developmental disabilities two daily living skills. In this study, the author worked with one youth (15 years) with DS. In the first stage, students were taught the skill of washing a table, whereas in the second stage they used the iPod Touch to teach themselves the vacuuming ability. The author reports that youth with DS and two participants were able to teach themselves the vacuuming skill with the support of the video prompting [11].

B. Robots and assistants for persons with DS

In the field of self-sufficiency support tools for persons with special needs, Alessi et al. developed the project named Casa+, a smart house aimed at supporting people with DS. This smart house offers functionalities in the following areas: domestic safety (guest authentication), time management (scheduling activities), daily actions assistance (aids for preparing food, taking care of the house, etc.), remote monitoring and control (for persons family/educators), among others. The authors do not report experimentation results [12].

On the other hand, Lehmann et al. explored the possibility of using robotic platforms in educational contexts with children with DS. To this aim, the authors introduced two robots to determine the childrens response during educational activities to develop social skills. The first robot is a child-size humanoid (Kaspar) whereas the second robot is similar to a small vehicle with cartoon-like features (Iromec). The authors worked with a girl with DS in three scenarios: a) dyadic and triadic interactions defined as collaboration and exchange, b) "move the robot" (improvement of body awareness), and c) imitation (ability to recognize a partial or complete sequence of movements). During the experiment, the girl played three months with each robot (once a week). Once the trial concluded, the results show that the girl prefers playing with the humanoid robot [13].

III. General system architecture

In order support the educational activities that teachers/special educators perform with children with DS, our tool performs the following stages (see the partial diagram of Figure 1):

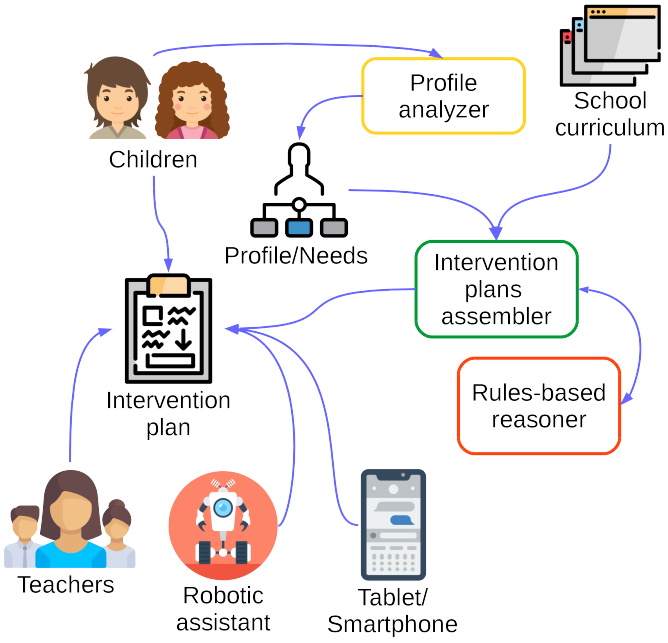


Fig. 1. A partial diagram of the system architecture.

- The *profile analyzer* analyses the most relevant characteristics of each child, and generates the data structure defined in Eq. 1:

$$\vec{P} = \{co_age, cr_age, ld_age, gender, level, be_results, \vec{CC}\} \quad (1)$$

Where:
- co_age represents the child's cognitive age (related with the intellectual quotient).

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:52:37 UTC from IEEE Xplore. Restrictions apply.

- cr_age is the chronological child's age whereas the $level$ is used to specify the year of primary education in which the child is currently enrolled.
- ld_age is the language developmental age of the child.
- $be_results$ are the results (scores) achieved by the child in the cognitive evaluation battery.
- $\vec{CC}$ is a set of elements ($\{cc_1, cc_2, \cdots, cc_N\}$) that represent the N educational curriculum contents of each $level$.

- With the aim of generating an *educational intervention plan*, the *intervention plans assembler* combines the data structure described above with the curriculum contents. In this stage the tool uses the *rules-based reasoning module* to determine which educational contents (described in the first section of this paper) must be addressed with the support of the robot and the mobile app.

Each of the rules of the expert system considers the different variables of the data structure mentioned above. In order to determine the level of difficulty for the mini-serious games (detailed later), the rules considers the results obtained by children in the evaluation battery, the existing gap between chronological and cognitive ages, the language developmental age and its difference with respect to the chronological age. Below we describe the Rule 1 that creates a fact representing the child's performance in the evaluation battery (considering a score between 0 and 10).

Rule 1.

```
@Rule (DifficultyLevel('be\_results'<<W()),
AND(TEST(lambda be\_results:be\_results>=0),
TEST(lambda be\_results:be\_results<=3)))
  ↳
Level(description="{The child's performance
falls in low-level.}")
```

Each plan that is automatically designed by the system contains several activities to be carried out with the support of the robotic assistant depicted in Figure 2. Some of the most relevant features of the robot are described below:

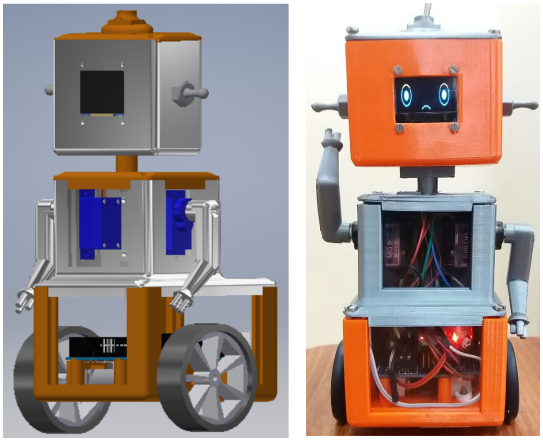


Fig. 2. The 3D design of the robotic assistant (left) and the assembled final version (right).

- Is a low-cost robot controlled by an Arduino One card and a Bluetooth card.
- Has 4 degrees of freedom (can move the two arms and can move in the $X-Y$ axis using two servo-motors).
- Includes an OLED i2c display that allows robot "expressing" emotions such as happiness, sadness, etc.
- Can be used in two modes: connected to a computer (through USB port) or with its built-in power source (4 hours of autonomy through two batteries of 3.7v and 2300mAh joined in series).

On the other hand, our system includes a mobile app (Figure 3) that can interact with the robot (through Bluetooth protocol). Some of the most relevant features of this app are the following:

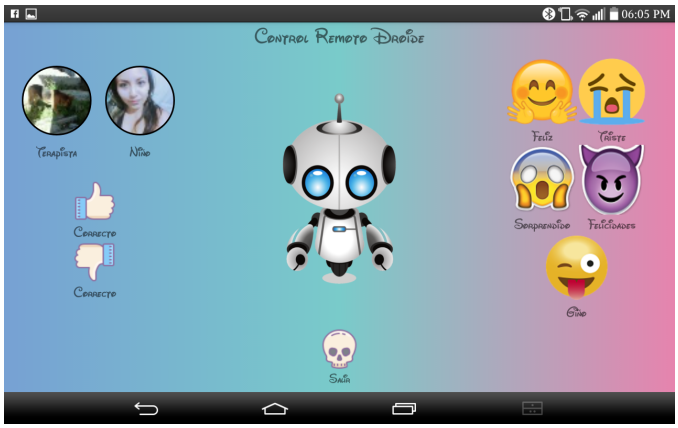


Fig. 3. A screen capture of the menu where children can control the robotic assistant and express emotions (angry, sad, happy, annoyed, etc.).

- Includes several mini serious-games organized in several levels (that are chosen according to children profile).
- Incorporates activities for all areas described in the introduction (speech-language therapy, attention, and concentration, body concepts and awareness, etc.).
- Provides several functionalities to support the decision-making for therapists/special educators (reports generation, children process monitoring, exercises/activities suggestions, etc.).

IV. Experiments and Results

With the aim of determining the children's response to the robot and the mobile application, we have conducted an experiment with 21 participants. Some of the most relevant aspects of the experiment are described below:

- The 21 participants are children aged between 6 and 11 years. Currently, they are enrolled in different school levels (from second to seventh).
- The participants are distributed according to the gender in two groups: 11 boys and 10 girls.
- The children not have disabilities and are included in normal educational programmes in their schools.

Each child used both the mobile app and the robotic assistant. This session consists typically on testing the most relevant

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:52:37 UTC from IEEE Xplore. Restrictions apply.

features of the system: the kinesthetic stimuli provided by the robot, the response of the mobile app during the execution of the serious games, and the interaction between a tablet and the robot. Once the working session finished, each child was asked to respond to a survey designed in the Likert scale. Figure 4 describes two of the most relevant aspects that we pretended to evaluate with the survey: children's perceptions about the robot and the mobile app. As it can be seen, the children feel very comfortable with the robot, and in the same way, they like the app.

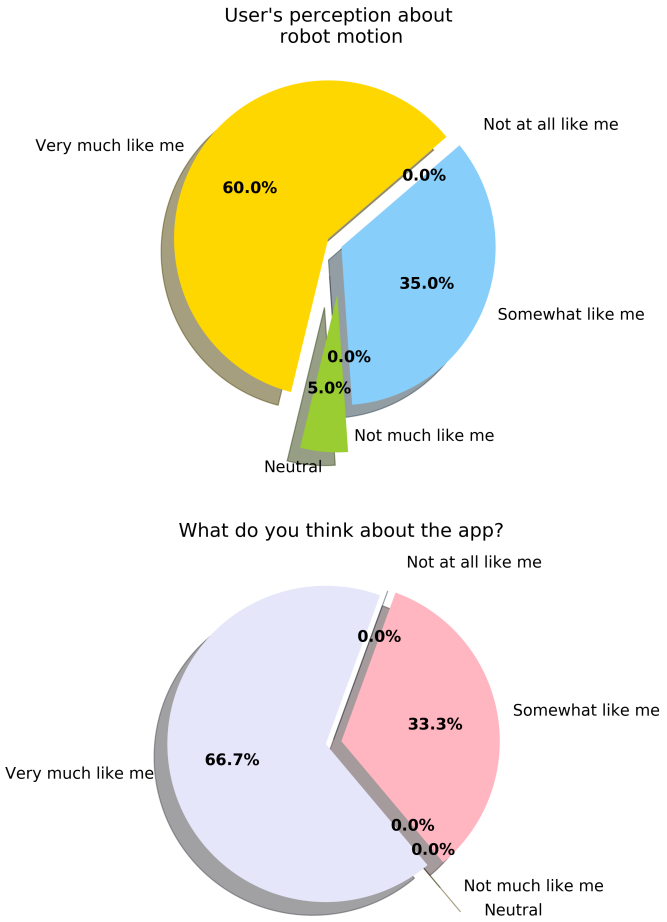


Fig. 4. General children's perceptions about the robot and the mobile app.

V. Conclusion

In this paper, we have presented the first evaluation stage of a support tool that relies on expert systems, robotic assistants, and mobile apps. In this line, it is important to mention that children have reacted in a very positive way to the robots appearance and functionalities. In the same way, the mobile application caught their attention (despite it is in an early development stage).

The robotic assistant can be easily reproduced given to the simplicity of the materials used for its construction. The final cost of the robot is around 50 USD (without considering the workforce).

As future work we propose the following activities:

- To conduct a perception analysis of the system with children with DS.
- To develop a feedback module to collect the therapists and children's perception of educational activities carried out with the support of the robot and the app.

Acknowledgment

The authors from the Universidad Politécnica Salesiana have been supported by the "Sistemas Inteligentes de Soporte a la Educación (v5)" research project, the Cátedra UNESCO "Tecnologías de apoyo para la Inclusión Educativa" initiative, and the Research Group on Artificial Intelligence and Assistive Technologies (GI-IATa).

References

[1] World Health Organization, "Genes and chromosomal diseases: Down syndrome," 2018.

[2] Misión Solidaria "Manuela Espejo", *Primer Estudio Biopsicosocial Clínico Genético de las Personas con Discapacidad*. Vicepresidencia de la República del Ecuador, 2012.

[3] M. B. Clavijo Barriga, "Lactancia materna y alimentación complementaria en relación al estado nutricional de niños y niñas menores a cuatro años de edad con Síndrome de Down que asisten al Centro Terapéutico Aprendiendo a Vivir en el periodo junio 2015," B.S. thesis, PUCE, 2015.

[4] J. Grieco, M. Pulsifer, K. Seligsohn, B. Skotko, and A. Schwartz, "Down syndrome: Cognitive and behavioral functioning across the lifespan," in *American Journal of Medical Genetics Part C: Seminars in Medical Genetics*, vol. 169, no. 2. Wiley Online Library, 2015, pp. 135–149.

[5] D. Conti, S. Di Nuovo, S. Buono, and A. Di Nuovo, "Robots in education and care of children with developmental disabilities: a study on acceptance by experienced and future professionals," *International Journal of Social Robotics*, vol. 9, no. 1, pp. 51–62, 2017.

[6] R. J. van den Heuvel, M. A. Lexis, and L. P. de Witte, "Robot zora in rehabilitation and special education for children with severe physical disabilities: a pilot study," *International journal of rehabilitation research. Internationale Zeitschrift fur Rehabilitationsforschung. Revue internationale de recherches de readaptation*, vol. 40, no. 4, p. 353, 2017.

[7] M. Gelsomini, M. Degiorgi, F. Garzotto, G. Leonardi, S. Penati, N. Ramuzat, J. Silvestri, and F. Clasadonte, "Designing a robot companion for children with neuro-developmental disorders," in *Proceedings of the 2017 Conference on Interaction Design and Children*. ACM, 2017, pp. 451–457.

[8] L. Kumin, "Resource guide to oral motor skill difficulties in children with down syndrome," 2015.

[9] I. Durango, A. Carrascosa, J. A. Gallud, and V. M. Penichet, "Using serious games to improve therapeutic goals in children with special needs," in *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services Adjunct*. ACM, 2015, pp. 743–749.

[10] S. Reis and A. M. Almeida, "The use of mobile devices to support daily routines of teens with down syndrome: a case study," in *Proceedings of the 7th International Conference on Software Development and Technologies for Enhancing Accessibility and Fighting Info-exclusion*. ACM, 2016, pp. 154–161.

[11] D. G. Brooks, "The effects of self-directed video prompting on teaching individuals with moderate to severe disabilities daily living skills," Ph.D. dissertation, The Ohio State University, 2012.

[12] R. Alesii, F. Graziosi, S. Marchesani, C. Rinaldi, M. Santic, and F. Tarquini, "Advanced solutions to support daily life of people affected by the down syndrome," in *Ambient Assisted Living*. Springer, 2015, pp. 233–244.

[13] H. Lehmann, I. Iacono, K. Dautenhahn, P. Marti, and B. Robins, "Robot companions for children with down syndrome: a case study," *Interaction Studies*, vol. 15, no. 1, pp. 99–112, 2014.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:52:37 UTC from IEEE Xplore. Restrictions apply.